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Editor-in-Chief and Editorial Team
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Dear Editor-in-Chief and Editorial Team,

Please consider our substantially rebuilt manuscript, “*When Learned IDS Model Selection Collapses to a Static Choice: A Measurement Study at the Edge*,” as a resubmission associated with prior manuscript TNSM-2026-10776.

Rather than restoring the earlier numerical claims, we rebuilt the data lineage, detector evaluation, scheduler implementation, and physical evidence. The study retains dataset-native pipelines for TON-IoT, CICIoT2023, and N-BaIoT. Its primary resource experiment comprises forty 500-second Raspberry Pi 4B runs in ten trace-matched blocks with raw USB-input power, telemetry, selected predictions, and decision logs retained for independent reconstruction.

The rebuilt evidence changes the conclusion. Relative to Static-MedRF, Tabular-Q saves 13.068 J per run (95% paired CI 9.736–16.400 J) and DQN saves 6.227 J (1.600–10.853 J), but both reduce F1 by 1.343 percentage points and recall by 2.954 percentage points and select LightLR in every formal window. We therefore do not claim a dynamic-switching advantage. A subsequent ten-block matched Static-LightLR control shows no resolved energy difference for Tabular-Q (+0.455 J, CI [−1.861, 2.772] J) and a +4.525 J DQN premium (2.693–6.357 J), with detection exactly identical to Static-LightLR. Isolated state-to-action timing is 10.8 μ s for Tabular-Q and 1.567 ms for DQN. Multi-seed, reward/state, and hyperparameter analyses further show that single-action behavior persists on observed held-out support; a detection-only reward changes the preferred static action to MedRF.

The revision adds two prospectively frozen physical controls. At fixed load, a confidence cascade escalates 1.34% of rows and recovers MedRF-like F1; we now show that its measured latency is specific to frequent small deferred-subbatch MedRF calls, not row-by-row escalation or a general cascade bound. Under a fixed 3×3 load/prevalence factorial, a predeclared DeadlineGuard switches between MedRF and LightLR. Relative to Static-MedRF it uses 43.714 J less per run (2.98%; 95% paired CI 41.418–46.010 J) and eliminates 512/5,000 deadline misses, at a 1.373-percentage-point F1 cost. The frozen DQN remains LightLR; frozen Tabular-Q instead enters unseen zero-initialized table rows whose deterministic first-maximum tie selects TinyDT, producing FPR 67.76%. We retain these unfavorable learned-policy results. A validation-only TinyDT threshold diagnostic explains much of its threshold-0.5 degradation but does not replace the primary result. A planned controller-only power arm is explicitly not executed because its frozen four-state subset was ambiguous; timing is not converted into joules.

We also derive a pairwise pre-deployment non-degeneracy test showing when the declared reward normalization cannot permit two actions to exchange rank on the observed support, and relate this failure mode to external-budget/SLO normalization.

The contribution is consequently measurement-grounded: learned IDS orchestration should be credited with adaptive benefit only after realized action occupancy, relevant static and conditional controls, deadlines, security loss, and whole-device cost are separated. The paper retains the single-device/KM003C and dataset-specific boundaries.

The evidence archives preserve invalid attempts and authorized replacements, close six executed-source hash gaps, and pass CRC, manifest, source-binding, and test-leakage checks. We confirm that the submission is original, is not under consideration elsewhere, and will be submitted with the approval of all authors.

Thank you for your consideration.

Sincerely,

Nuonan Ouyang
on behalf of all authors